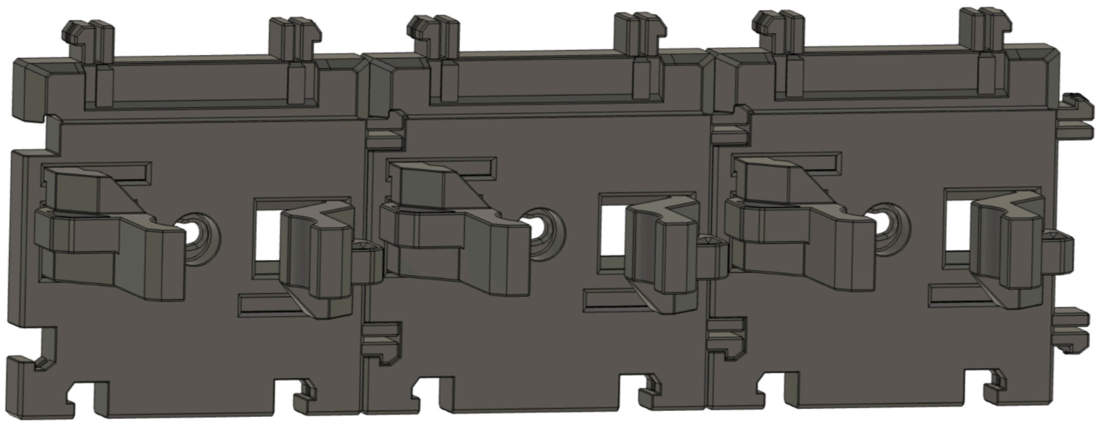
CARGO CONNECT: GRIPPER



User Guide

Resources, Print Settings and Assembly

Before You Start..

What's Included

The below files and documents are included within your download:

1. **Ready to Print (.3mf file) -** All the parts in one place and ready to print on Bambu Lab printers.
2. **User Guide (.pdf file) -** A complete 'User Guide' to help you print and assemble the model.
3. **Individual parts (.stl files) -** All the 3D printable parts you need to make the complete base model, compatible with most slicers.
4. **Assembly View Only file (Do Not Print) -** This file is to view the fully assembled model only and not for printing.
5. **Optional Parts (.zip folder) -** A folder containing alternative versions of the Cargo Connect: Gripper, such as parts with varying tolerances.

What We Used

Printers:

Bambu Lab P1S

You can get the printers we used here (affiliate link)* **- Bambu Lab P1S**

Filaments:

Gripper - Bambu Lab PLA Basic Dark Grey

You can get the filaments we used here *(affiliate link*)* **- Bambu Lab Filament**

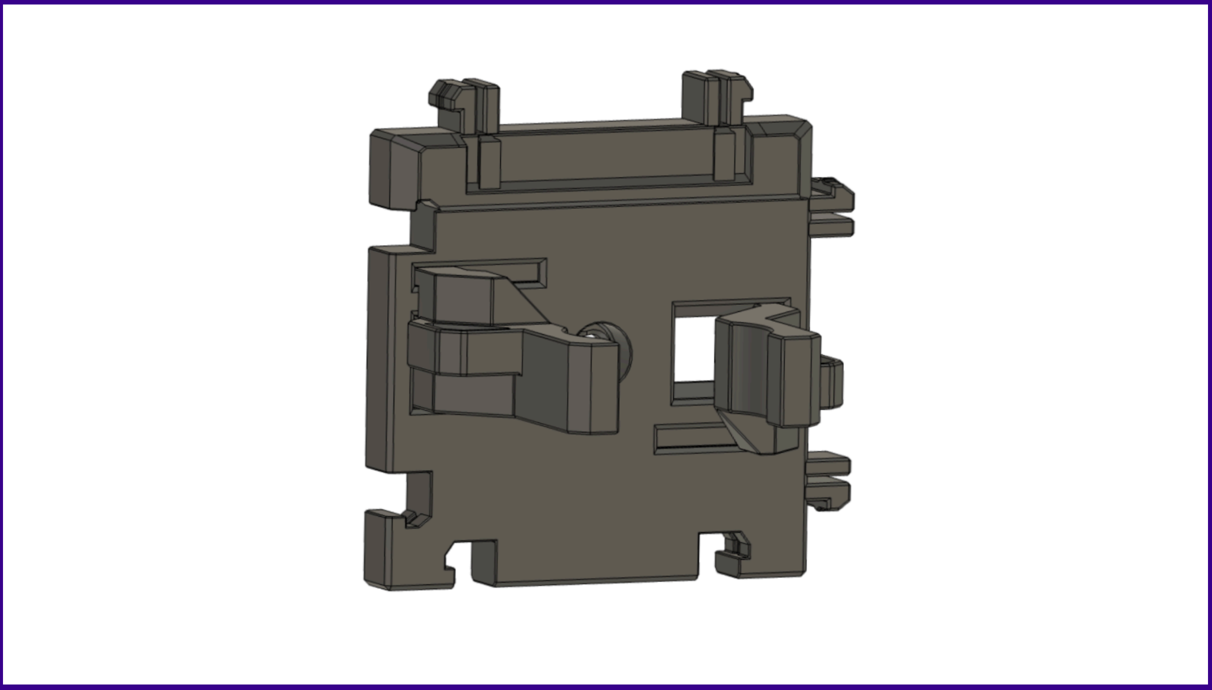
*Some links in this document are affiliate links, which means we may earn commission if you purchase through them.

Cargo Connect: Gripper - Variations

The Cargo Connect: Gripper comes in two options listed below for ultimate flexibility.

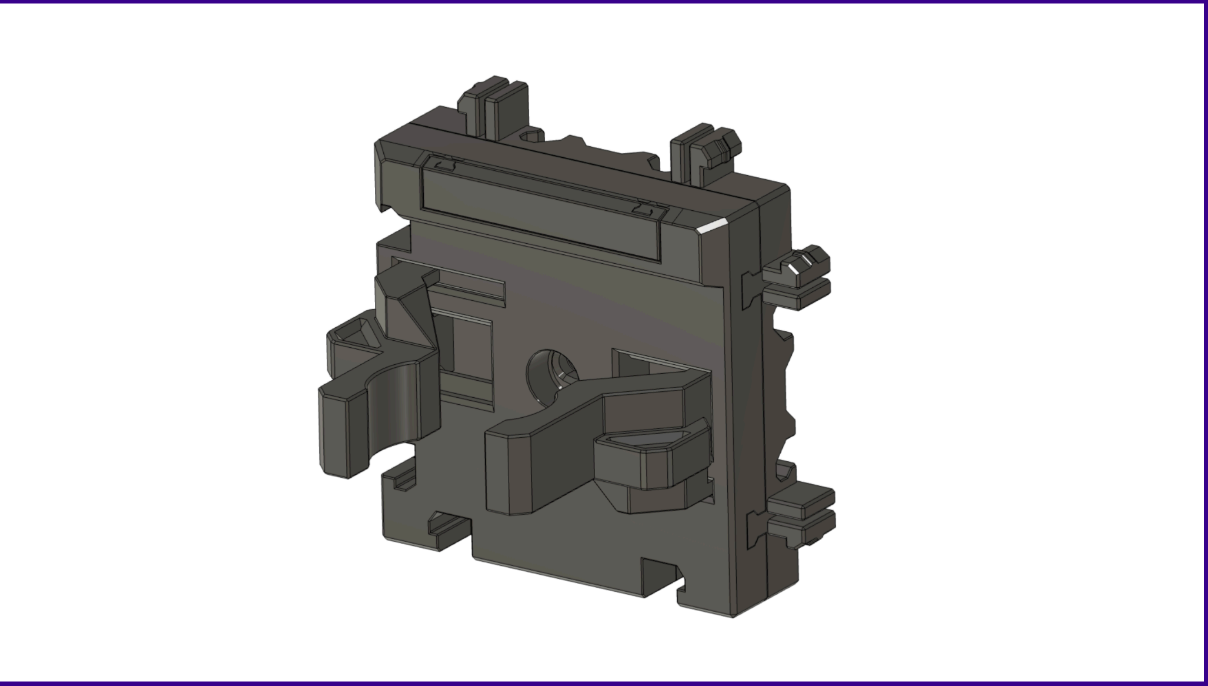
Fixed Gripper 1x1

A single tile with the Cargo Connect interlocking system around each side. This tile is perfect when just simple, fixed location Grippers are all that's needed.



Sliding Gripper 1x1

The same adjustable Gripper, but with a custom mounting plate designed to slide along the Cargo Connect: Rails - horizontal and vertical. The mounting plate still includes the interlocking system around each side, making this the perfect tile for ultimate flexibility on the Cargo Modular System.



Print Settings

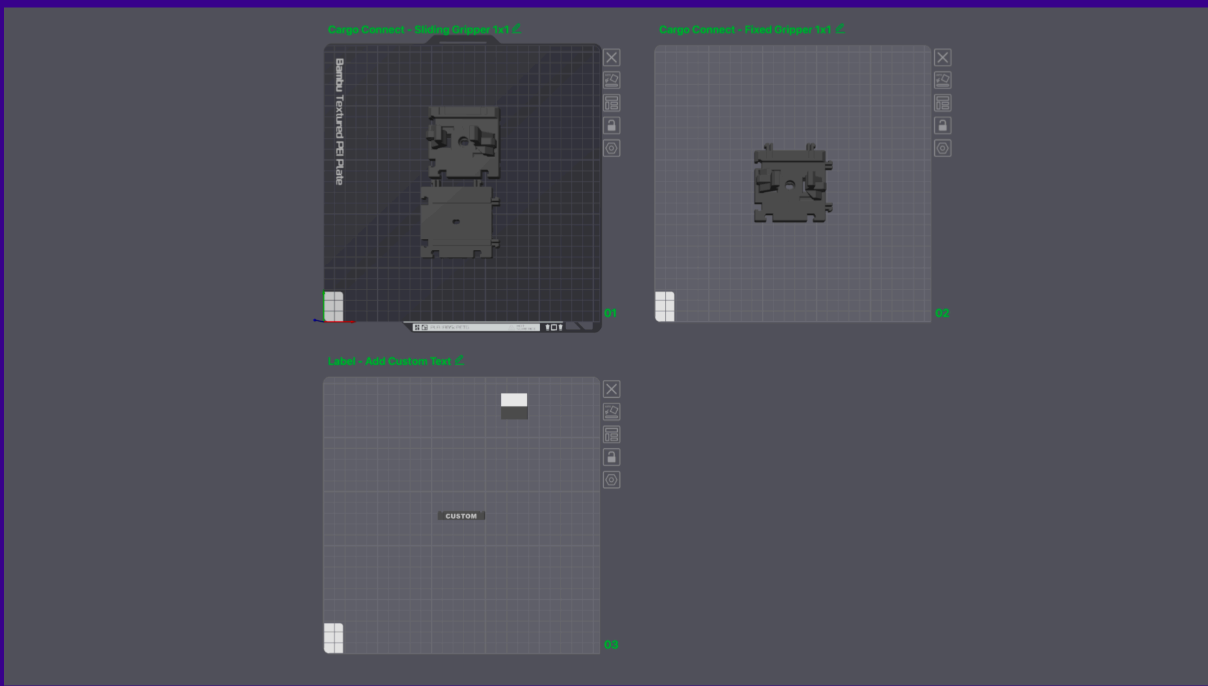
Important Note:
The information detailed in this ‘Print Settings’ document are the settings we found gave the best results for us. Your printer, materials and printing conditions are unique to you, and due to these widely varying aspects you will know what settings work best for your printers.

Ready to Print Project

A .3mf file named **‘1.Ready to Print’** is included in the download. This file is a pre-prepared Bambu Studio project, which includes the settings we used.

Both ‘Fixed Gripper - 1x1’ and ‘Sliding Gripper - 1x1’ have been included in the 1. Ready to Print project, so you can simply slice the variation you want to use.

(As mentioned above, these settings worked best for us and for our needs, however, you may need to adjust these settings to suit your printing setup and needs)



Individual Part Settings

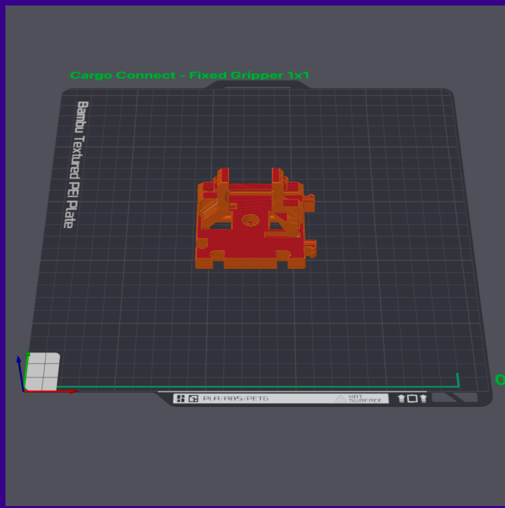
Fixed Gripper Tile

Basic Settings (Non-Standard):

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 3
- **Infill** - 25%
- **Supports** - None
- **Brim** - Auto

Estimations:

- **Print Time** - 1hr 9m
- **Filament Used** - 32.86g



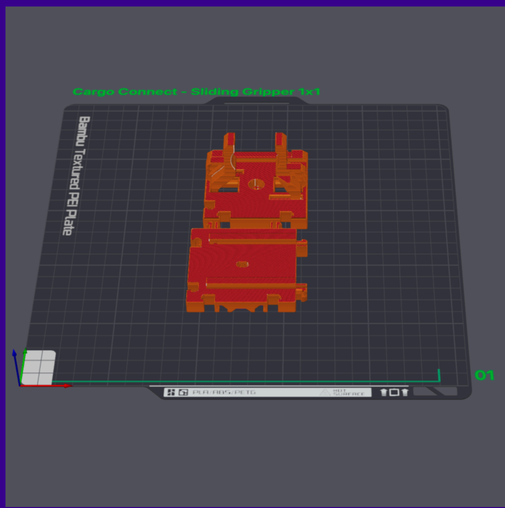
Sliding Gripper Tile

Basic Settings (Non-Standard):

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 3
- **Infill** - 25%
- **Supports** - None
- **Brim** - Auto

Estimations:

- **Print Time** - 1hr 56m
- **Filament Used** - 58.81g



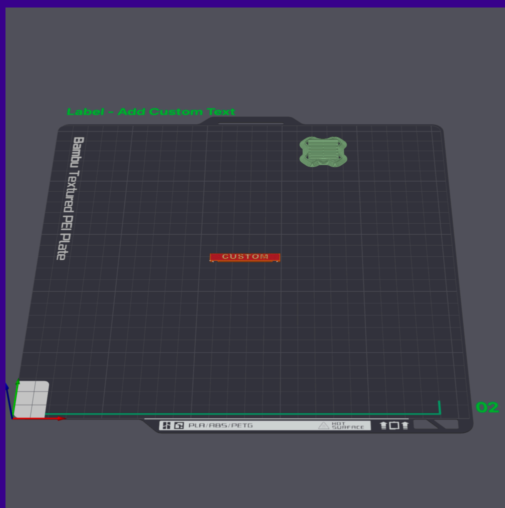
Labels *(To make 'Custom Labels', see step 5)*

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer** Walls - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto

Estimations:

- **Print Time** - 5m 41s
- **Filament Used** - 2.11g



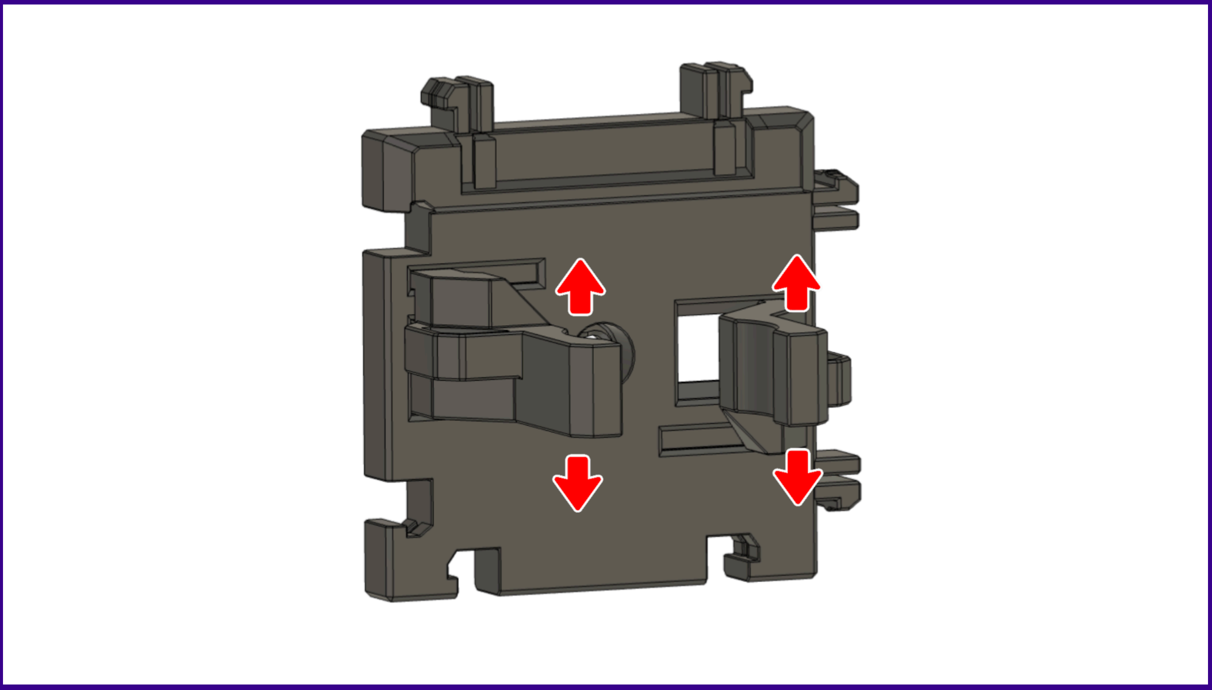
Assembly Guide

Before Assembly:

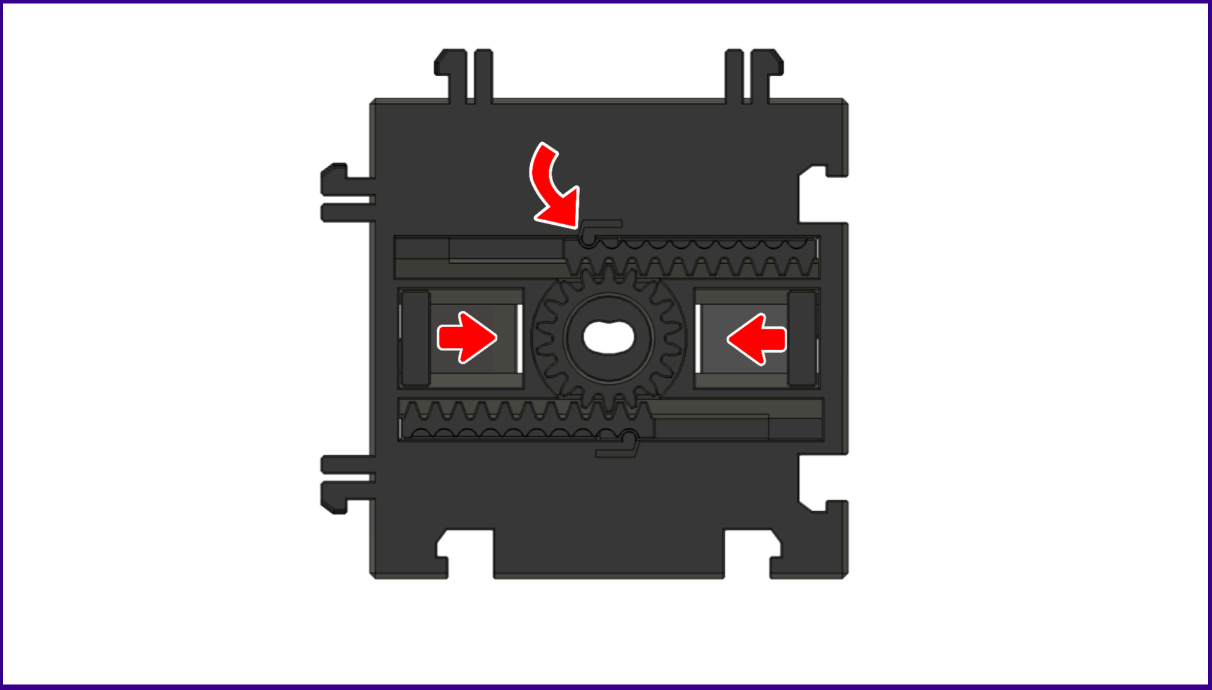
Once all parts are printed, remove any supports and check for any imperfections or poorly printed areas - *these may need post processing work or reprinting.*

Step 1 - Preparation

After printing and before use, gently wiggle each gripper arm gently back and forth. This will help to release the rack and pinion mechanism on the back and allow it to move freely. (This step may not be needed, depending on your printer's tolerances).



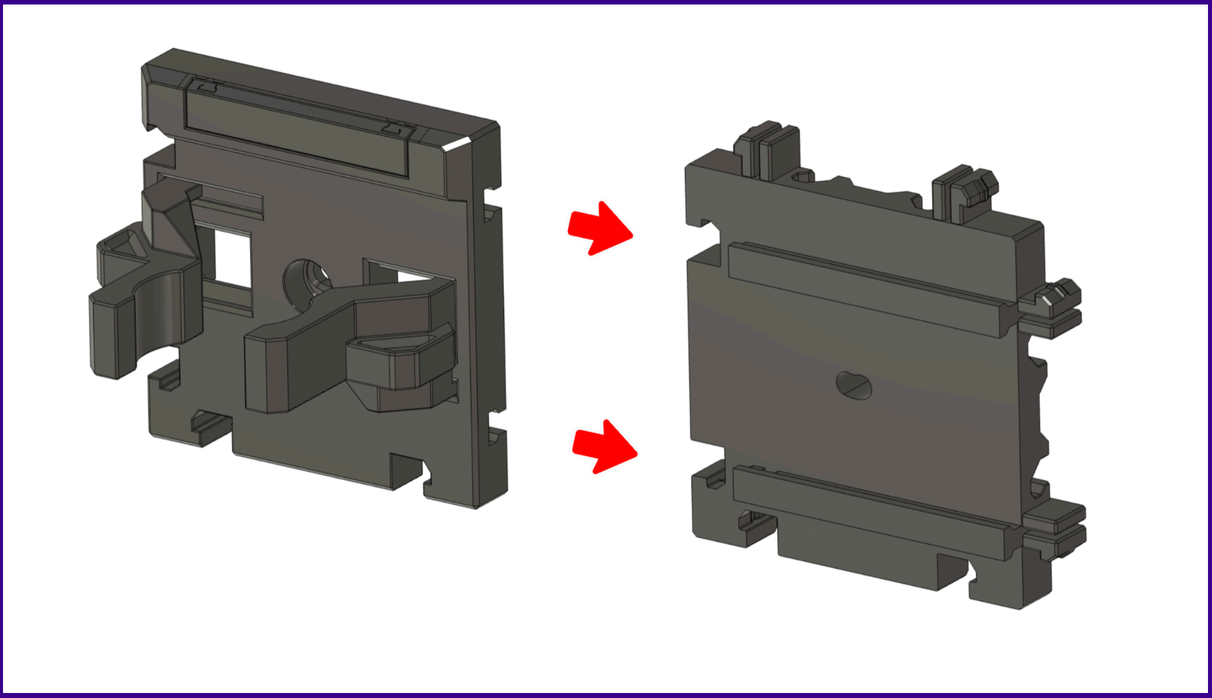
Test the Gripper by moving the arms back and forth to check they move freely, then by gripping an item to ensure the tab and groove locking mechanism shown below is working securely.



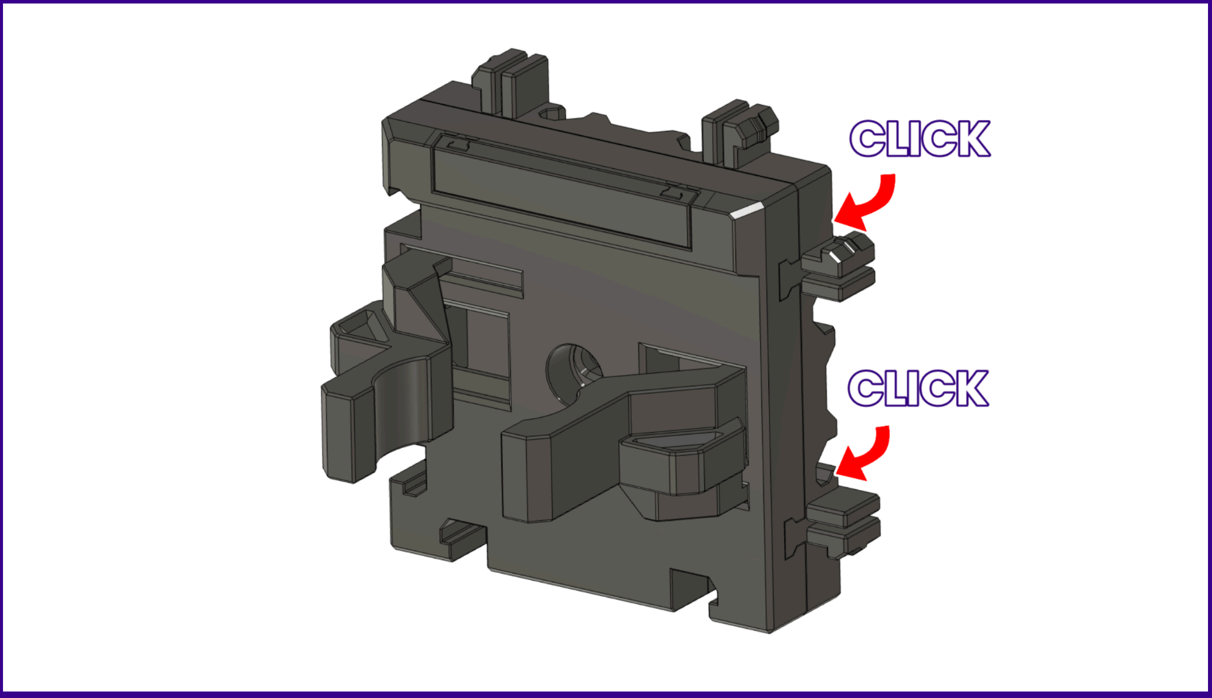
Step 1 - continued..

If you're using the Sliding Gripper tile, you will need to attach the two parts of the print together before use. Simply slide the top tile onto the rails on the mounting plate as shown below (We recommend testing the Gripper function as above before connecting the parts together).

Once fully engaged you should hear a click and the tiles should be perfectly aligned on all sides.

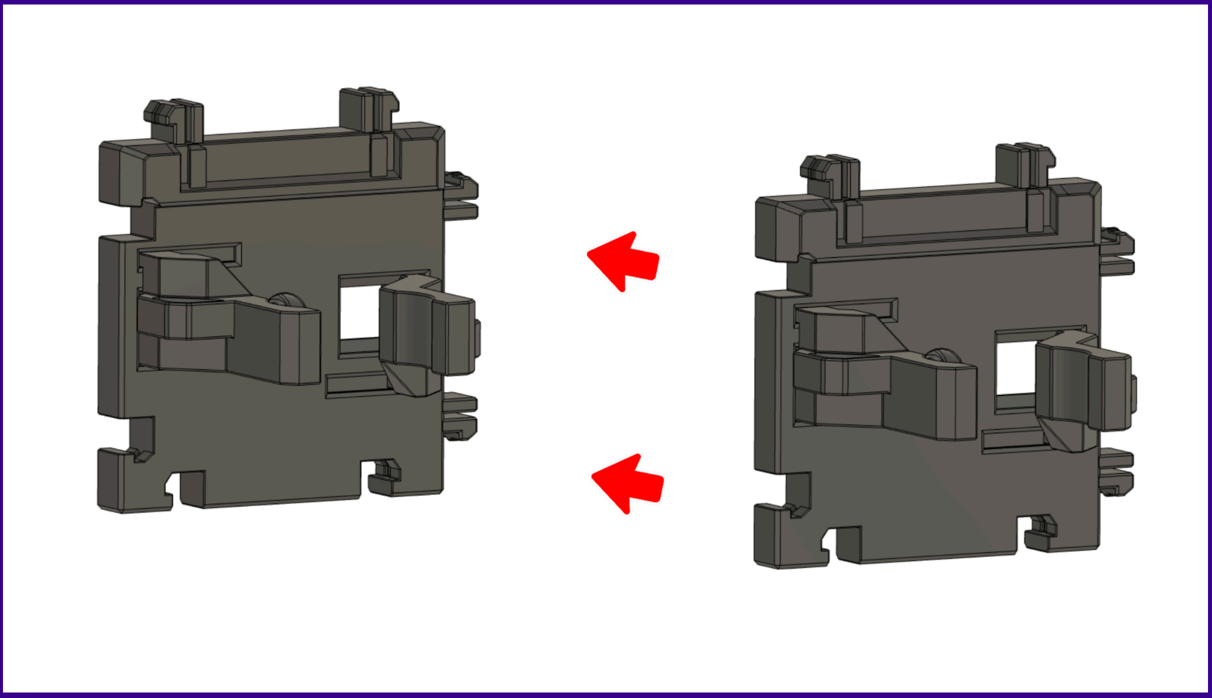


The two parts are designed to fit together securely and should not feel loose or come apart once assembled. They are **not** intended to be separated after clicking together. If the fit is too loose or the parts slide apart, we recommend using the **'Sliding Hooks – Tighter Tolerance'** version found in the **Optional Parts** folder instead. These parts provide a firmer connection and should resolve any fitting issues.

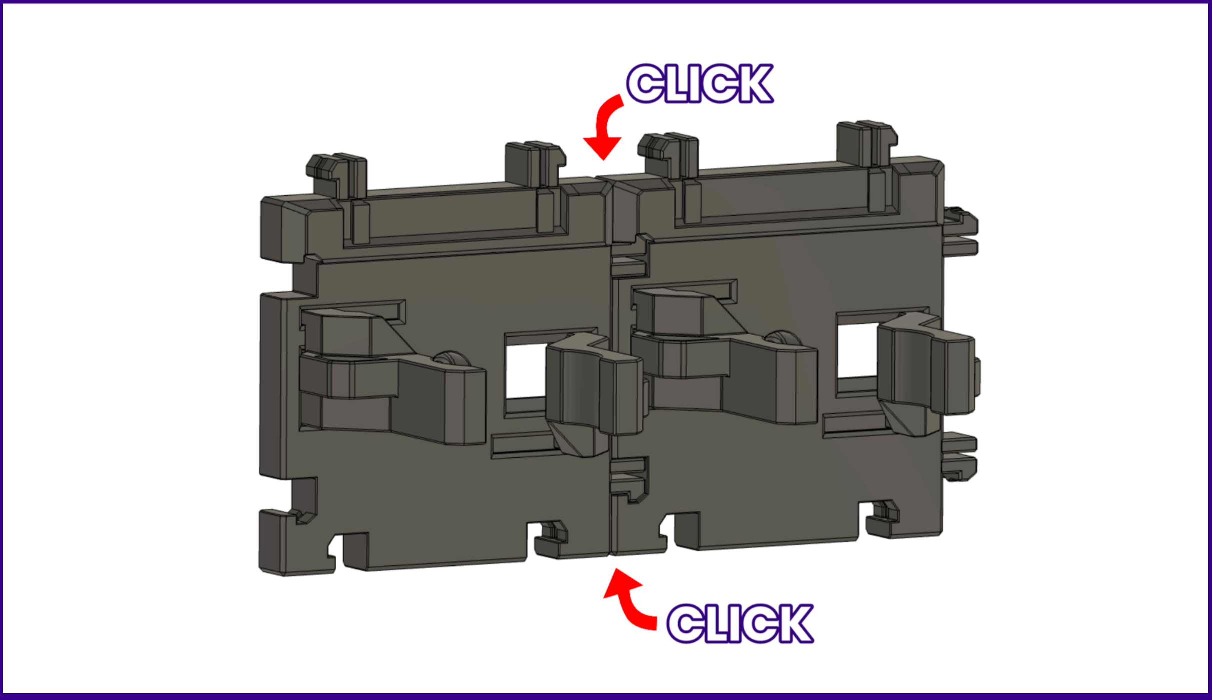


Step 2 – Connecting Multiple Tiles

Each tile (Fixed or Sliding version) can connect to another tile on any of the four sides using the Cargo Connect Interlocking System - in a 'Jigsaw' like connection. *(The tiles should be orientated with the label slot at the top).*



Lay the tiles on a hard flat surface and push them together until you hear the parts click together.

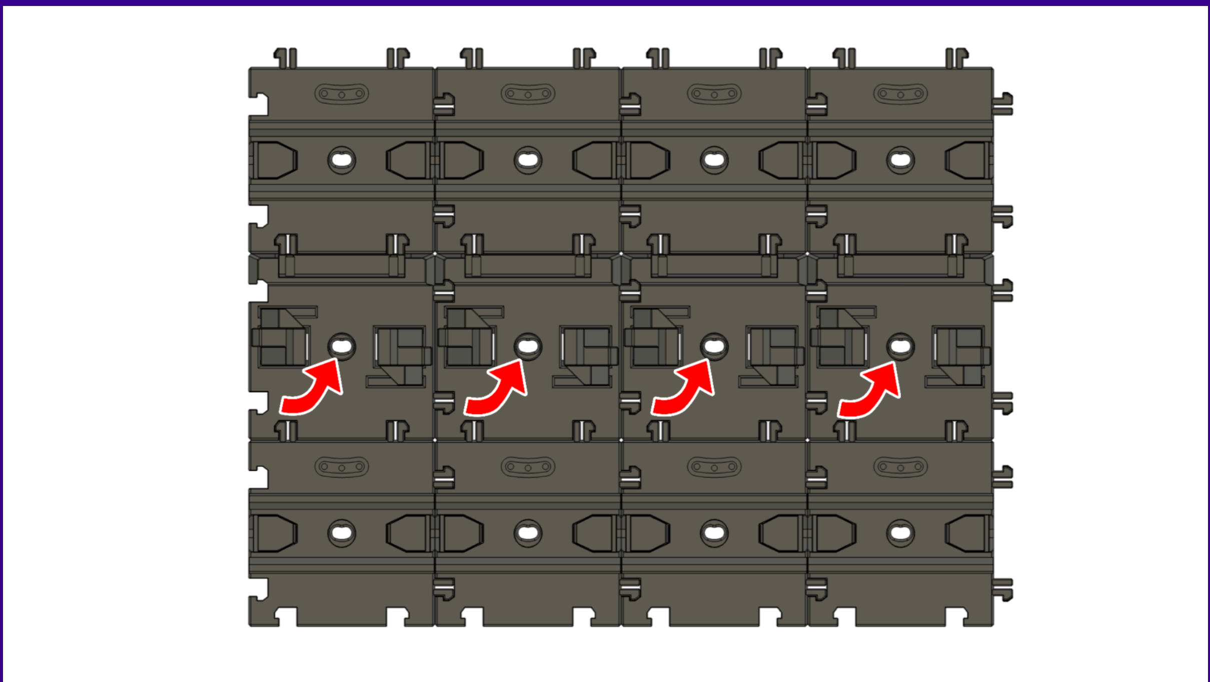


Step 3 - Wall Mounting

Mounting the 'Fixed Gripper' tile

Each tile includes a screw hole for wall mounting. While the interlocking design provides inherent stability, we recommend adding an appropriate screw fixing to each tile, as items held by the grippers will add extra stress to the built in interlocking design.

As each tile only has one screw hole, you will need to attach at least two tiles together and use the screw hole on each to ensure the tiles do not rotate when in use.



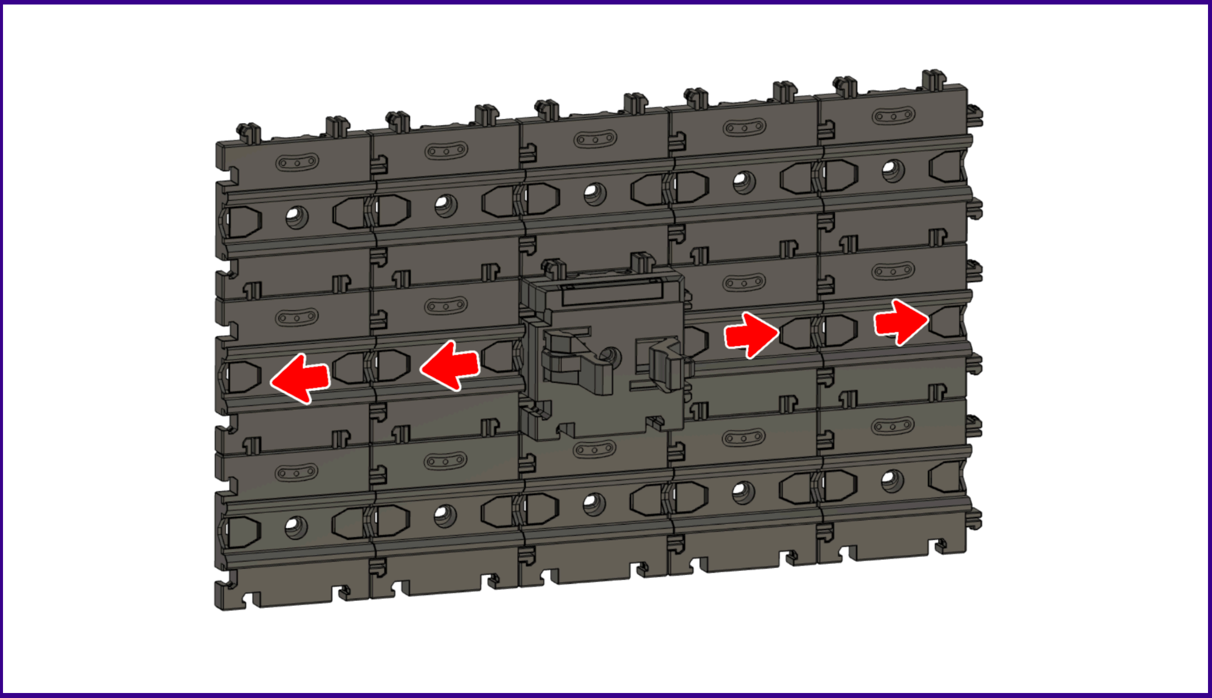
Important Note

The number and placement of tiles and screws used for the Cargo Modular System is **entirely at your discretion,** subject to the strength of your wall and the intended weight of your stored items. **You should thoroughly test the stability and strength of your wall and the system before use and inspect the system at regular intervals to ensure the materials used have not deteriorated.**

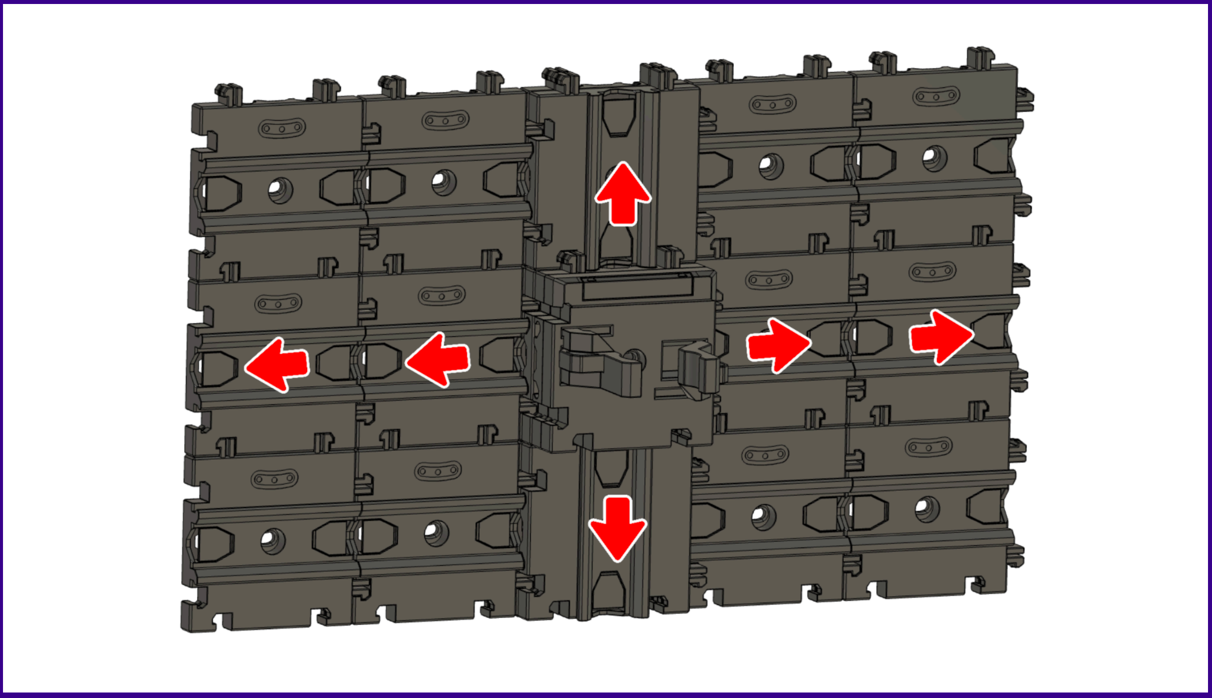
Mounting the Sliding Gripper tile

The Sliding Gripper tile, once assembled as per Step 1, has tracks cutout on the back that allow it to slide along the Cargo Connect: Rails. The Sliding Gripper is also fully compatible with the stoppers built into the Cargo Connect: Rails.

Horizontal Rails:



Vertical Rails:

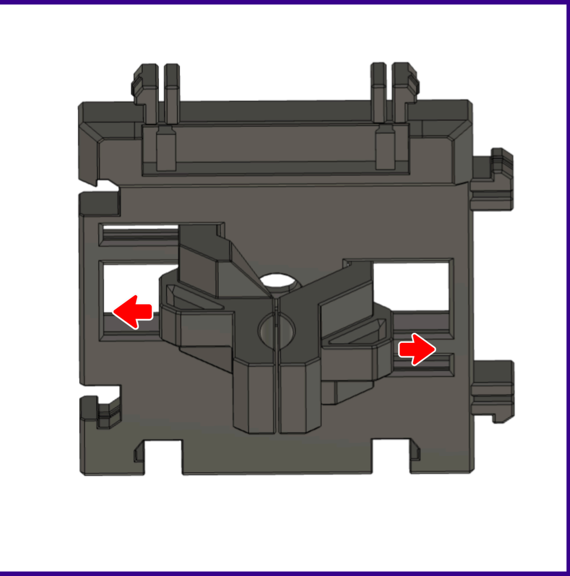


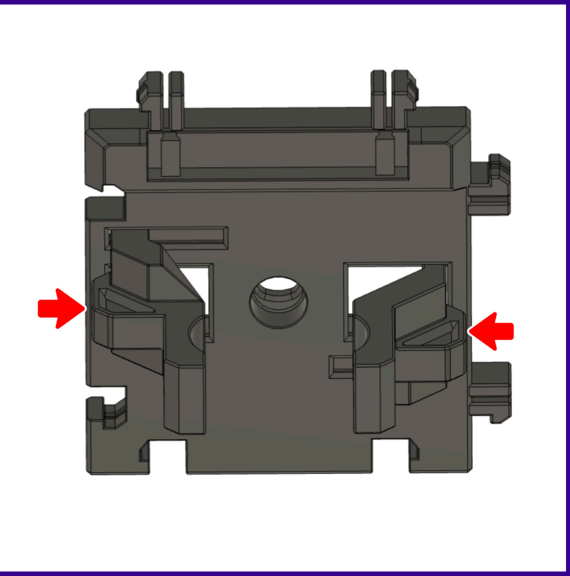
Step 4 - Use

The gripper arms automatically lock to your desired size using a tab and groove mechanism built into the tile. To operate the arms, simply follow the steps below:

- **To open to grippers** - use the two handles on the side of each arm and pull outwards.

- **To close the grippers** - push the two arms together simultaneously from the base of the handles (Do not open and close the grippers using the top of the arms).





Important Note

The gripper arms have been thickened to improve strength when holding items, but please remember that FDM 3D printed parts have inherent strength limitations. Always consider the weight, shape, and size of the items you store, and ensure they're suitable for the printed part.

We recommend carefully testing each item in the gripper before regular use to confirm stability and fit, and checking at regular intervals for any signs of material fatigue or deterioration over time.

Always ensure tools are securely seated and not easily dislodged before leaving the system unattended.

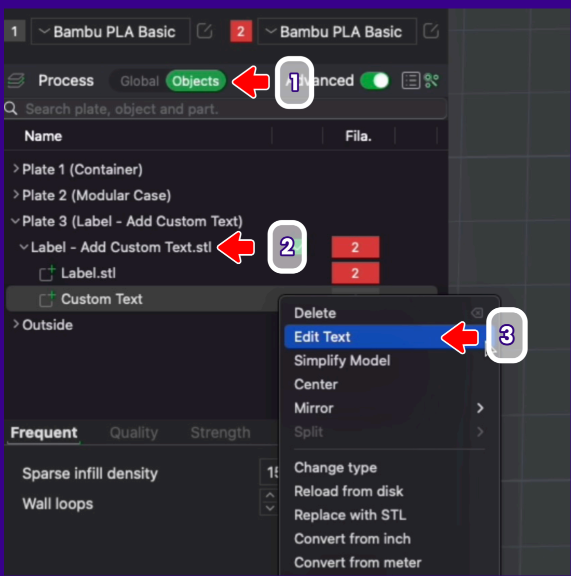
Increasing the infill percentage and number of wall/perimeter lines in your slicer settings may help improve durability, but results will vary depending on your printer and material.

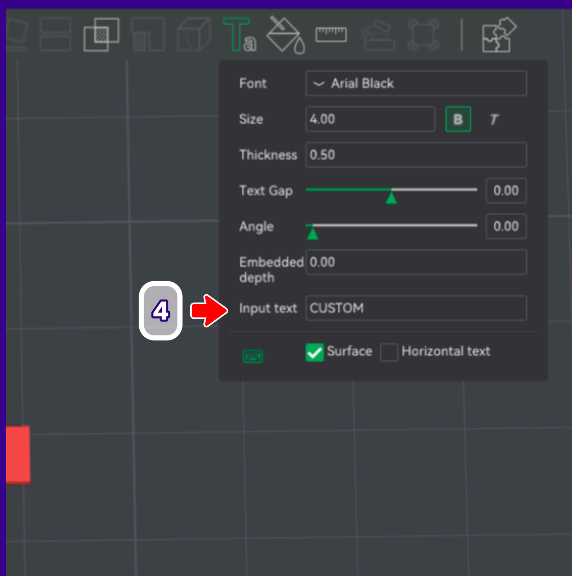
Step 5 - Custom Labels

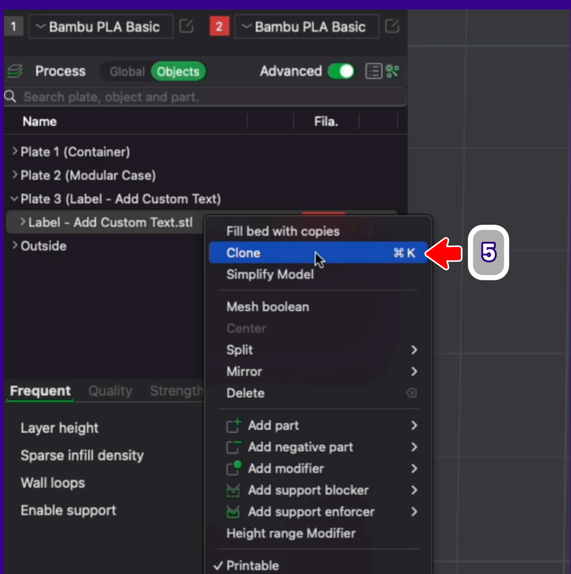
There is a label slot on the top of each Gripper tile. We have included a preset editable label inside the '1. Ready to Print' Bambu Studio .3mf file. To alter the text in Bambu Studio, follow the steps below:

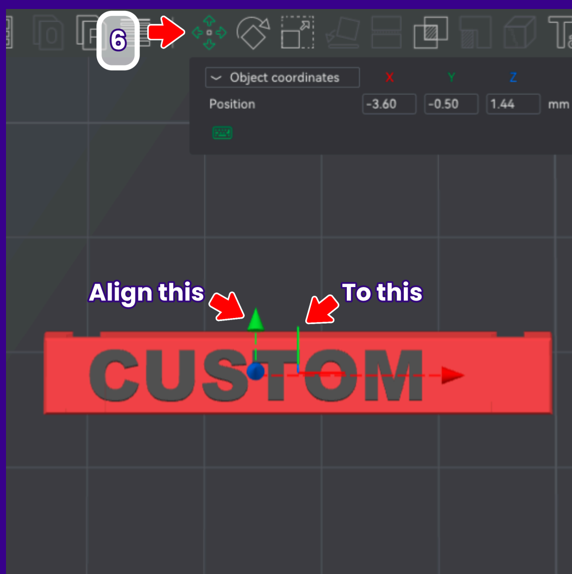
(There is also a 'Blank Label.stl' included for use with other slicers).

1. Navigate to the 'Objects' tab
2. Click on the drop down for the 'Label - Add Custom Text' STL file
3. Right click on the 'Custom Text' object and select 'Edit Text'
4. In the 'Text Box', you can change the text using the 'Input Text' box. *(You can also edit the size, font etc during this process)*
5. If you want to make numerous labels, you can right click on the whole 'Label - Add Custom Text' STL and select clone, then choose how many copies you need and follow the process above for each label.
6. The Custom Label object has already been centered, however when you alter the text, depending on the letters you use, it will likely un-center the text horizontally. If you select the 'Custom Text' part and select 'move', you can realign the green lines shown in the picture and this should re-center the text.



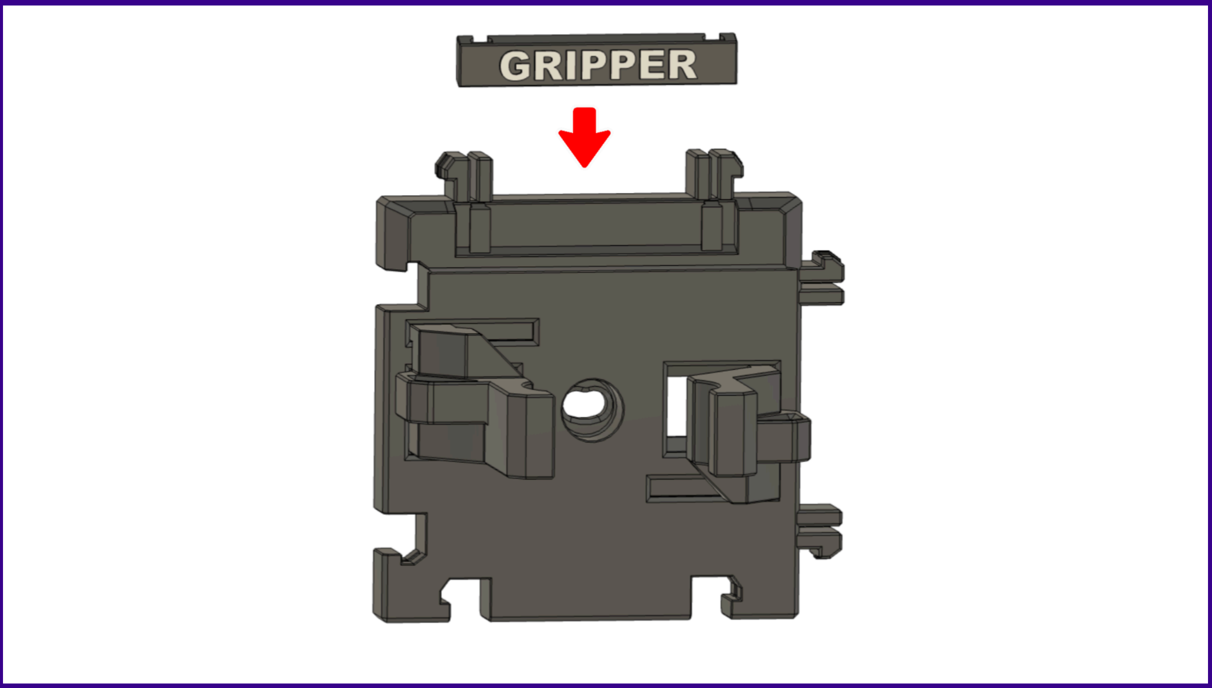


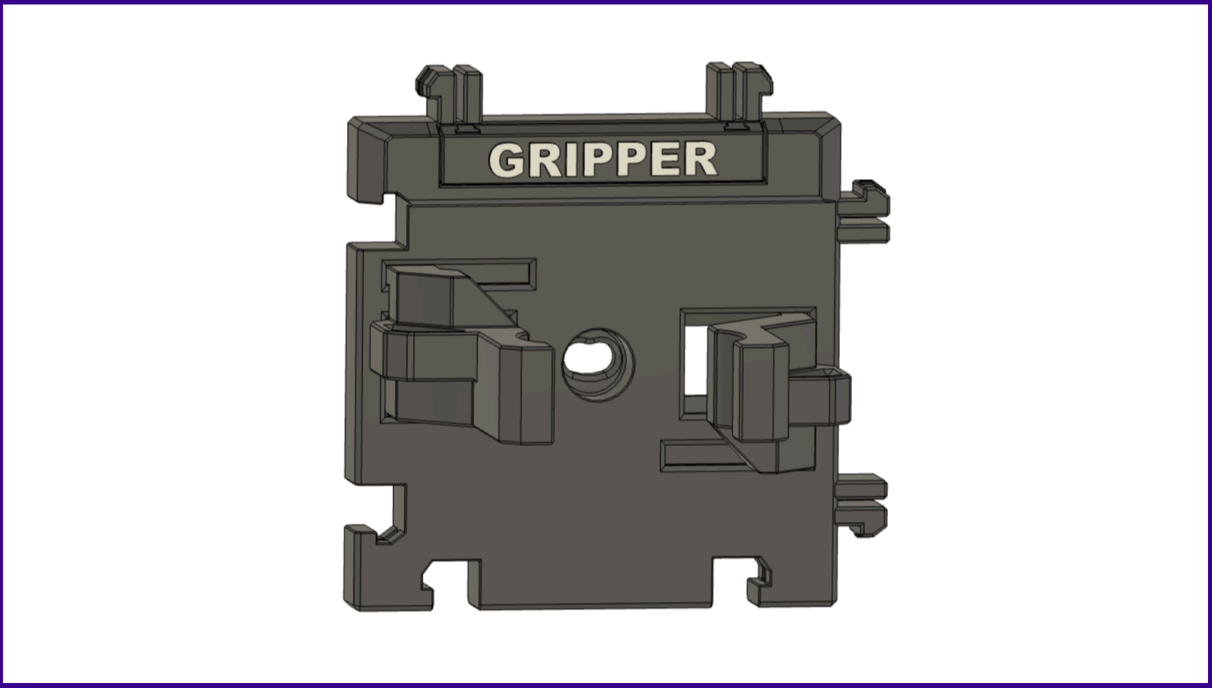




Step 5 *continued..*

Once you have printed the labels, they slide into place on top of each tile.

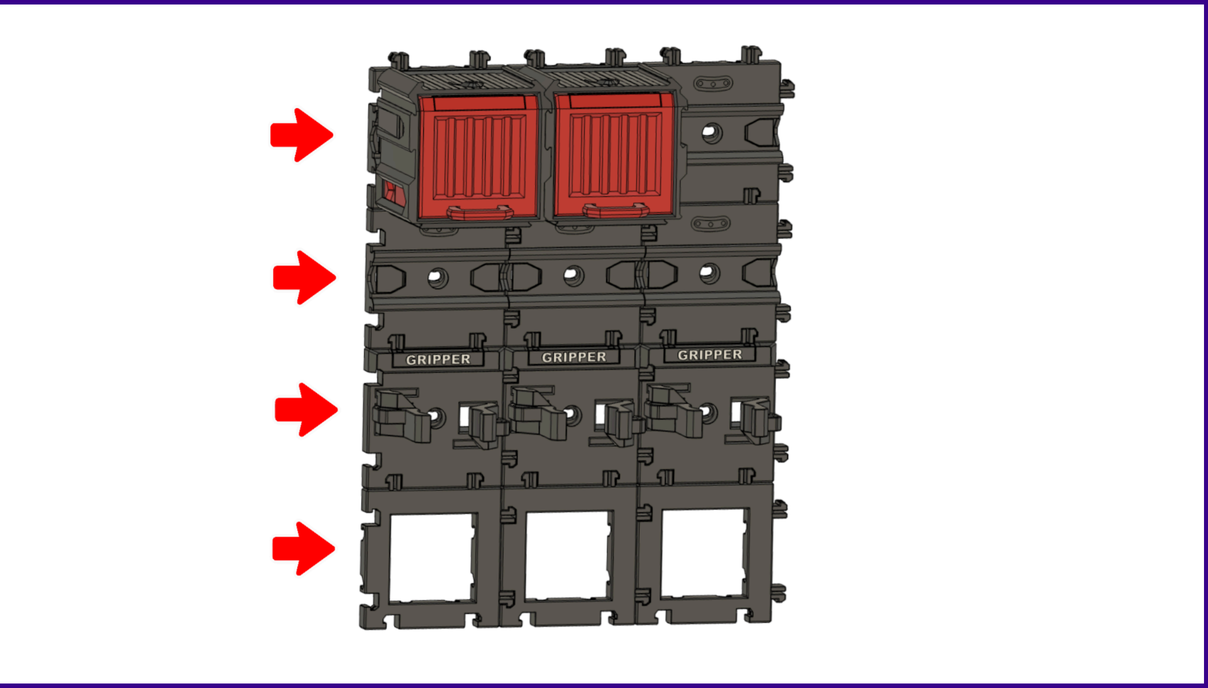




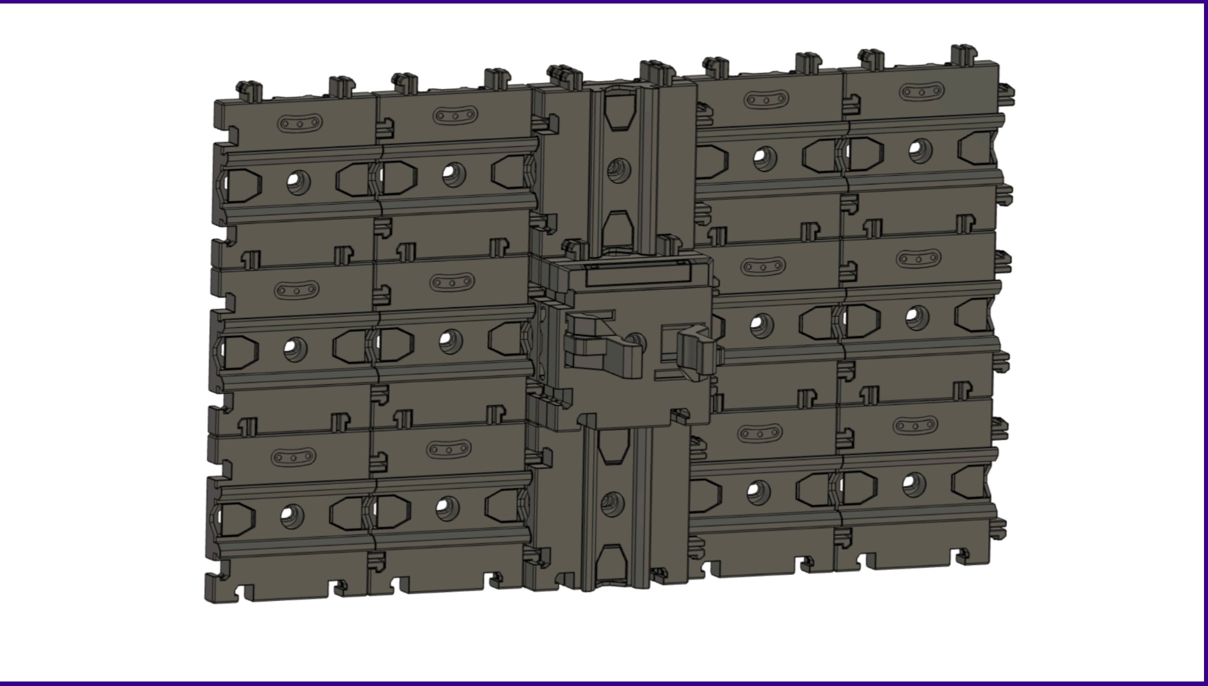
Compatible Components

The Cargo Connect: Gripper can interlock with all other Cargo Connect tiles, meaning you can build the perfect storage system for all your needs.

Simply use the interlocking pegs around the sides of each tile to connect your desired tiles together.



The 'Sliding Gripper' tile is also fully compatible with the Cargo Connect: Rails, refer to Step 3 for full information.



Troubleshooting

We put a lot of time into making our designs as easy to print and use as possible, testing extensively with various filaments and printers. That said, we understand that occasional issues can occur - below are some potential solutions that may be useful:

Warping and Bed Adhesion

Warping is a common challenge in FDM 3D printing, often caused by poor bed adhesion, rapid cooling, or inconsistent ambient temperatures.

It's important that tiles and components come off the bed as flat as possible, as warping may impact features like interlocking pegs, sliding fits, or hinge alignment. If you notice lifting corners or uneven bases, try adding a brim or applying a suitable adhesive directly to the build plate to improve bed adhesion.

Ensuring consistent ambient temperatures and proper first-layer calibration can also go a long way in reducing warping across prints.

Tolerances

The Cargo Modular System is designed to be as print-friendly as possible, and we've worked hard to make tolerances as forgiving as possible across a wide range of printers. However, the unique features and functionality of our designs can rely on tight tolerances to work as intended — especially for parts that slide, hinge, or interlock.

If you're experiencing issues with fit or movement, it's worth checking your printer calibration — including flow rate, retraction settings, and nozzle temperature — to ensure accurate extrusion. Also make sure your filament is dry and of consistent quality, and that your slicer profile is correctly set up for your printer and material.

If problems still persist, we recommend printing a tolerance test model to help identify where your setup might need fine-tuning. These are widely available from reputable sources such as Thangs.

Use Disclaimer

Every Play Conveyor model has been designed and tested with care, but results will vary depending on your printer, materials, settings, and environment. Because of this, we cannot guarantee print outcomes across all setups.

You are responsible for ensuring that both the printing and assembly of any models are completed to a safe and functional standard. Always test your prints — especially components that are wall-mounted, load-bearing, or move — and check them regularly for wear or failure.

This user guide is provided as a general reference to support your build, but it is not exhaustive or guaranteed to suit all applications. Use your judgment where needed to suit your specific setup.

All files are provided "as is" and without warranty. Play Conveyor Ltd accepts no responsibility for any injury, damage, or loss resulting from the use or misuse of our designs.

This disclaimer highlights some of the key points when using our models, but it does not replace or override the official 'License Agreement'.

The 'License Agreement' associated with your subscription tier is the official document that defines your permitted use of the model, any restrictions, and your responsibilities. Please ensure you review the relevant license agreement to confirm proper usage.

Thangs Members

- Maker tier: covered by the **Personal License** (for personal use only)
- Merchant tier: covered by the **Commercial License** (includes permitted commercial use)

Your applicable license can be found under each model's listing on our Thangs page.

Patreon Members

- The license for all Patreon 'Maker Access' tiers is included within the downloadable files for each design.

It is your responsibility to review and follow the terms of your applicable license.